

Optimal Droop Setting for Congestion Reduction in a 100% Grid-Forming Inverter-based Power System

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Abstract— The high penetration of inverter-based resources (IBRs) introduces new challenges to power systems due to the complex inverter control. However, IBRs can be configured to maximize their benefits to improve system resilience and reliability. This paper proposes a steady-state optimization model that aims to mitigate transmission congestion in a 100% grid-forming (GFM) IBR-based power system. This goal is achieved by determining the optimal droop settings for the GFM IBRs under different congestion conditions due to renewable energy and load variations. The numerical solution is rigorously verified by a high-fidelity model of the IEEE 39-bus test system with detailed GFM IBR control in the time-domain electromagnetic transient (EMT) simulation tool PSCAD. The numerical solution and simulation results show a significant congestion reduction while meeting all other operating requirements. It is also observed that the numerical solving time is substantially less compared to the EMT simulation time.

Index Terms - grid-forming inverter, transmission congestion, optimal power flow, computational methods

I. INTRODUCTION

To realize decarbonization goal, the penetration level of inverter-based resources (IBRs) to integrate renewable energy to power systems continues to increase quickly around the world. With advanced control methods, this introduction of IBRs allows them to provide additional control capabilities and valuable grid services, which improves system reliability and resilience. At the device level, significant amounts of work focus on developing IBR models and control technologies [1], [2]. At the system level, substantial efforts have also been dedicated to study the impacts, mitigate potential consequences, and create guidances and grid-code requirements to realize higher IBR penetration and maximize their benefits [3], [4].

IBR control can be classified into grid-following (GFL) and grid-forming (GFM) technologies. While the former lets an IBR behave as a controllable current source, the latter allows an IBR to operate as a controllable voltage source behind an impedance. Most of existing IBRs in the current practice are deployed with GFL control for the main goal of providing maximum available power to the system, without or with a small curtailment for reserve requirements. However, GFM IBRs have been gaining significant momentum because more and more of their benefits are being realized. For example, compared to GFL IBRs, GFM IBRs allow faster transient responses to regulate voltage and frequency [5]. Droop-based

and virtual synchronous machine (VSM) controls are the two most popular GFM control methods, and their in-depth comparisons are discussed in [6].

As part of the work toward a future grid with high IBR-based penetration, several recent works have been focusing on a power system with 100% IBRs. In [7], potential challenges from both economic and technical perspectives of reaching 100% IBRs are discussed. In [8], dynamic simulations of the planning 2031 Western Electricity Coordinating Council (WECC) interconnection system with 100% droop-based GFM IBRs are conducted. The study shows promising transient stability characteristic offered by GFM IBRs under several severe contingencies. In [9], a wide-area oscillation analysis of the WECC system with up to 100% GFM IBRs is discussed. The phasor-domain simulation results show that high GFM IBR penetration levels introduce new oscillation modes, compared to those triggered by the existing synchronous generators.

While all of the aforementioned works are related to transient stability analyses, several research has focused on steady-state modeling of a power system with high IBR penetration. A formulation is proposed in [10] to determine the optimal droop coefficients for GFL inverters in distribution feeders. A power flow model and solution method for a microgrid with droop-based distributed generation is proposed in [11]. At transmission level, a sequential power flow solution framework for a system with both conventional synchronous generators and IBRs is developed in [12]. This framework integrates the steady-state model of GFM IBRs into the governor power flow feature in the commercial transient stability simulation tool PSS/E. These system-level works, however, have not addressed the capability of GFM IBRs in improving system performance. With the use of droop coefficients in droop-based and VSM-based GFM control, it is also possible to determine the optimal droop settings for the GFM IBRs to maximize their benefits.

To aid to the existing work on steady-state modeling and optimizing GFM IBRs, this paper proposes an optimization-based model and solution verification to mitigate transmission congestion due to thermal limits in an extreme power system with 100% droop-based GFM IBRs and optimal droop settings. Unlike the conventional approach of setting droop coefficients based purely on generator ratings, the proposed method also considers transmission constraints when choosing the values for the droop coefficients with the use of an accurate power flow model. To demonstrate the feasibility and efficacy of the obtained optimal solution, a high-fidelity electromagnetic transient (EMT) simulation in PSCAD is conducted.

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II. PROPOSED FORMULATION FOR OPTIMIZING DROOP SETTINGS OF GFM IBRS

This section describes the proposed optimal droop setting problem in the rectangular coordinates. Several definitions and assumptions about the system are given as follows:

- The sets of all n buses, d load buses, and m GFM IBR buses are denoted by $\mathcal{N}$, $\mathcal{D}$, and $\mathcal{M}$, respectively.
- The sets of all b branches are denoted by $\mathcal{B}$.
- Among n buses, one angle-reference bus is chosen as the first GFM IBR bus. The imaginary component f_1 of the complex voltage at this bus is equal to zero.

A. Objective function

The objective of minimizing the total overloaded power of transmission lines is given in the following function:

$$\min f(\mathbf{X}) = \sum_{(i,k) \in \mathcal{B}} \Delta P_{ik}, \quad (1)$$

where ΔP_{ik} is the overloaded power of line (i, k) :

$$\Delta P_{ik} = \begin{cases} P_{ik} - P_{ik}^{max} & \text{if } P_{ik} > P_{ik}^{max}, \\ 0 & \text{otherwise.} \end{cases} \quad (2)$$

B. Variables

The variable vector includes real and imaginary bus voltage components, system frequency ω , GFM real and reactive power generation, and GFM $P-f$ droop coefficients:

$$\mathbf{X} = [e_1, \dots, e_n, f_2, \dots, f_n, \omega, P_1^g, \dots, P_m^g, Q_1^g, \dots, Q_m^g, m_1^P, \dots, m_m^P]. \quad (3)$$

C. Constraints

1) *Nodal power balance equations*: The set of nodal power balance equations $\mathbf{g}_i^P$ and $\mathbf{g}_i^Q$ are given by:

$$g_i^P = P_i^{inj} - P_i^g + P_i^l = 0, \quad (4)$$

$$g_i^Q = Q_i^{inj} - Q_i^g + Q_i^l - Q_i^{sh} = 0, \quad \forall i \in \mathcal{N}, \quad (5)$$

where P_i^l , Q_i^l , and Q_i^{sh} are the real and reactive power from loads and shunt capacitors. The power P_i^{inj} and Q_i^{inj} injected into the system at the i^{th} bus are calculated as follows:

$$P_i^{inj} = \mathbf{G}_i \cdot (e_i \mathbf{e} + f_i \mathbf{f}) + \mathbf{B}_i \cdot (f_i \mathbf{e} - e_i \mathbf{f}), \quad (6)$$

$$Q_i^{inj} = \mathbf{G}_i \cdot (f_i \mathbf{e} - e_i \mathbf{f}) - \mathbf{B}_i \cdot (e_i \mathbf{e} + f_i \mathbf{f}), \quad \forall i \in \mathcal{N}, \quad (7)$$

where $\mathbf{G}_i$ and $\mathbf{B}_i$ are the i^{th} rows in the conductance and susceptance matrices $\mathbf{G}$ and $\mathbf{B}$.

2) *Power generation from GFM IBRs*: The real and reactive power generation P_i^g and Q_i^g in (4) and (5), respectively, of GFM IBR at bus i are calculated as follows.

$$\omega = \omega^* - m_i^P (P_i^g - P_i^*) \frac{S_{base}}{M_{base}}, \quad \forall i \in \mathcal{M} \quad (8)$$

$$V_i = V_i^* - m_i^Q (Q_i^g - Q_i^*) \frac{S_{base}}{M_{base}}, \quad \forall i \in \mathcal{M}, \quad (9)$$

where Q_i^* is usually set to zero. Because of the $P-f$ and $Q-V$ droops, all 4 local variables at this bus, i.e. e_i , f_i , P_i^g , and Q_i^g , are unknown, compared to just two unknown variables as in conventional power flow. The system frequency ω , which

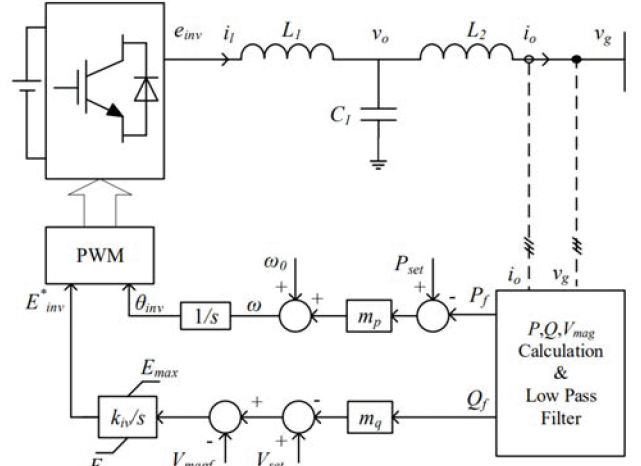


Fig. 1. Single-loop droop-based control for GFM inverters [5]

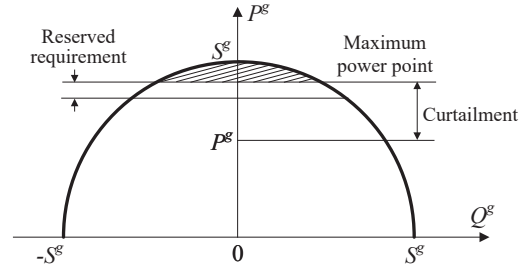


Fig. 2. Power capability curve of IBRs with reserve requirement and curtailment of real power generation.

can be considered as a global variable for the entire system, is also unknown prior to solving power flow. It is important to note that droop coefficients m_i^P and m_i^Q , which are defined with respect to IBR rating M_{base} , need to be converted to system base S_{base} in order to be incorporated in the power flow equations (4) and (5). In this proposed problem, the $P-f$ droop coefficient in (8) is constrained as follows:

$$\underline{m}^P \leq m_i^P \leq \bar{m}^P, \quad \forall i \in \mathcal{M}. \quad (10)$$

3) *PQ capability limit of an IBR*: The real and reactive power generation of an GFM IBR need to satisfy the PQ power capability limit as follows:

$$(P_i^g)^2 + (Q_i^g)^2 \leq (\bar{S}_k^g)^2, \quad \forall i \in \mathcal{M}. \quad (11)$$

This constraint is derived from the current limiting requirement of the IBR [13], and it is represented by the outer half circle in Fig. 2.

4) *Reserve requirement for IBRs*: As also shown in Fig. 2, the real power generation of a GFM IBR needs to satisfy the spinning reserve requirement specified by system operators for fast frequency response [4]:

$$0 \leq P_i^g \leq (1 - \alpha) \bar{P}_i^g, \quad \forall i \in \mathcal{M}, \quad (12)$$

where α is the reserve requirement coefficient.

5) *Voltage constraints at load buses*:

$$\underline{V}^2 \leq e_i^2 + f_i^2 \leq \bar{V}^2, \quad \forall i \in \mathcal{D}, \quad (13)$$

where $\underline{V}$ and $\bar{V}$ are the lower and upper limits of the voltage at load buses.

6) Frequency constraints:

$$\underline{\omega} \leq \omega \leq \bar{\omega}, \quad (14)$$

where $\underline{\omega}$ and $\bar{\omega}$ are the required lower and upper limits of the system frequency.

7) *Transmission line flow limits:* The non-negative variable ΔP_{ik} in the objective function (1) is introduced to relax the original hard constraint on the flow P_{ij} of line (i, k) as follows:

$$-(P_{ik}^{max} + \Delta P_{ik}) \leq P_{ij} \leq (P_{ik}^{max} + \Delta P_{ik}), \quad (15)$$

$$\Delta P_{ik} \geq 0, \quad (16)$$

where line flow P_{ik} is calculated as follows:

$$P_{ik} = g_{ik}(e_i^2 + f_i^2 - e_i e_k - f_i f_k) + b_{ik}(e_i f_k - f_i e_k), \quad \forall (i, k) \in \mathcal{B}. \quad (17)$$

D. General Form of the OPF Formulation

Considering the objective and all constraints described above, the proposed optimization problem can be given in the following general form:

$$\min f(\mathbf{X}) \quad (18a)$$

$$\text{s.t. } \mathbf{g}(\mathbf{X}) = \mathbf{0}, \quad (18b)$$

$$\underline{\mathbf{h}} \leq \mathbf{h}(\mathbf{X}) \leq \bar{\mathbf{h}}, \quad (18c)$$

$$\underline{\mathbf{X}} \leq \hat{\mathbf{X}} \leq \bar{\mathbf{X}}, \quad (18d)$$

In the above formulation, equality constraint $\mathbf{g}(\mathbf{X})$ in (18b) is constructed from the power balance equations (4)-(7) and droop equations (8)-(9). The functional inequality constraints of variables in (11), (13), (15), and (17) are represented by $\mathbf{h}(\mathbf{X})$ in (18c). In (18d), $\hat{\mathbf{X}}$ is a sub-vector of $\mathbf{X}$ that contains the variables that are imposed by box constraints (10), (12), (14), and (16).

Due to the nonconvex feasible set resulting from constraint (13), the optimization problem (18) is nonlinear nonconvex. Among available methods in the literature, the predictor-corrector interior-point algorithm has been showing good performances in solving such a nonlinear nonconvex problem. Therefore, in this work, the framework in [13]–[15] based on this algorithm is adopted to solve (18).

III. VERIFICATION APPROACH IN EMT TOOL PSCAD

To verify the numerical solution after solving the formulated optimization problem in Section II, it is necessary to verify the feasibility of the solution and its efficacy in improving system performance, i.e. reducing transmission congestion in this work. A powerful verification approach is using the time-domain EMT simulation tool PSCAD. Moving from the steady-state domain to a dynamic simulation, the following requirements are performed as preparation steps for the verification of the numerical solution:

- Same system configuration and parameters such as line and transformer parameters, GFM IBR ratings, GFM droop coefficients, and load are applied in the optimization input data and PSCAD model.
- The detailed switching model and control shown in Fig. 1 of a single-loop droop-based GFM IBR is developed and verified separately in PSCAD.

IV. CASE STUDY

This section describes the case study, assumptions to generate congestion data, and the optimal droop coefficients and its efficacy in reducing overloaded transmission in the system. The comparison between the numerical solution and EMT simulation in PSCAD is also included.

A. Description of the Test System and Congestion

Fig. 3 shows the modified IEEE 39-bus system used as the test system and solution evaluation. The IEEE 39 bus system, also known as the 10-machine New England power system includes 10 synchronous generators, 46 transmission lines and 19 loads with a combined 6,116 MW demand [16], [17]. In this work, the synchronous generators at Buses 30 to 38 are replaced by droop-based GFM IBRs with the same rating and initial dispatch. On the other hand, the synchronous generator at Bus 39, which was created to represent the power transfer between this area and the nearby area, is replaced by an equivalent negative load. Other power flow data of the modified system is similar to that in the original system. The key system parameters related to the formulated optimization in Section II is shown in Table I.

There is no available data regarding thermal-loading line limits for the IEEE 39-bus system. Therefore, to create congestion cases for evaluating the efficacy of the optimal droop coefficients, the power flow solution of the modified IEEE

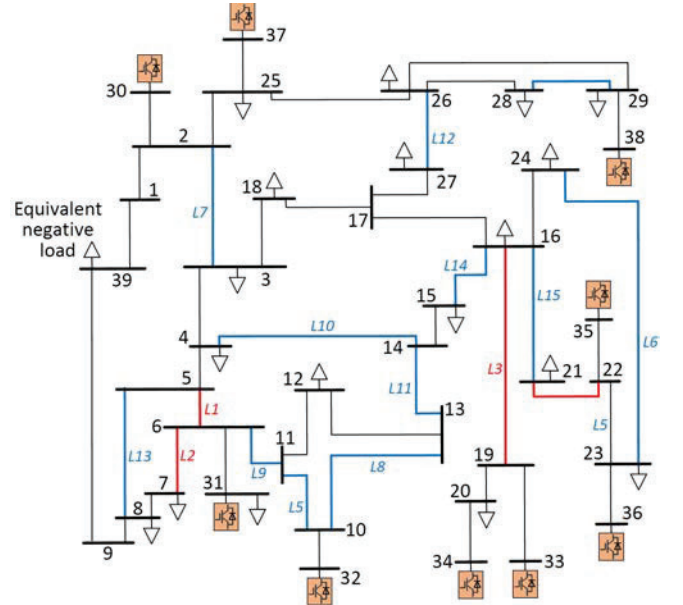


Fig. 3. The IEEE 39-bus system with all synchronous machines being replaced by GFM IBRs. Overloaded lines are highlighted in red (Case 1) and in red and blue (Case 2).

TABLE I
KEY PARAMETERS USED IN THE CASE STUDY

Parameter Descriptions	Values
$P - f$ droop coefficient limits (m^P and $\bar{m}^P$)	1% - 10%
Same $Q - V$ droop coefficient m^Q for all IBRs	5%
Load voltage limits ($\underline{V}$ and $\bar{V}$)	0.9 pu - 1.1 pu
Frequency limits ($\underline{\omega}$ and $\bar{\omega}$)	59.7 Hz - 60.3 Hz
Reserve requirement (α)	5%
Convergence tolerance for optimization solution	0.001 pu (0.1 MVA)

TABLE II
CREATION OF TWO CONGESTION CASES

Line flow being considered as overloaded	Resulting number of overloaded lines	Overloaded range
Case 1: ≥ 300 MW	4	11 % - 25 %
Case 2: ≥ 200 MW	15	5 % - 25 %

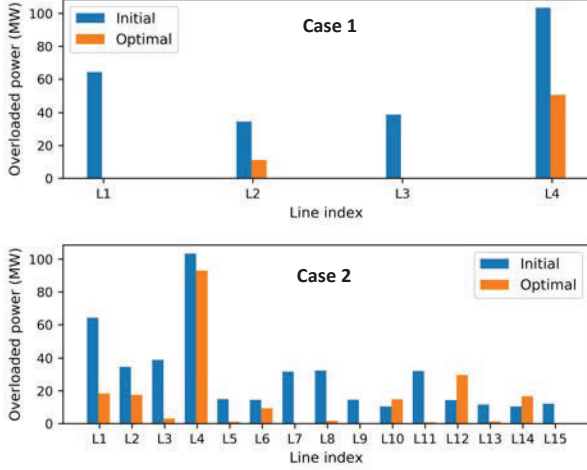


Fig. 4. Comparison of individual line flows without and with optimal droop settings in Cases 1 and 2.

39-bus system is used. To determine the maximum limit, or P_{ik}^{max} in (15), several selected transmission lines with the highest power flow are multiplied by multipliers ranging from 0.8 to 0.95. Other lines with substantially lower power flow are assumed to have sufficient headrooms; thus, they are not constrained by (15). Table III shows the detailed line flow-based criterion to select overloaded lines in two studied cases with different congestion levels.

B. Efficacy of the Optimal Droop in Reducing Congestion

Fig. 4 shows the comparison of individual line flows in the selected transmission lines with line limit constraints in the two studied cases. It can be seen that the congestion reduction is substantial, and the overloading condition in several lines is eliminated. In addition, since there are initially more overloaded lines in Case 2 while the number of decision variables, i.e. m_i^P , is unchanged, the levels of overloading condition of Lines L1-L4 in Case 2 are less than those in Case 1. In Case 2, there are few lines such as L12 that experience higher overloaded power. However, the total overloaded power, which is the objective of the optimization problem, reduces significantly from 241 MW to 62 MW in Case 1 and 441 MW to 207 MW in Case 2, as shown in Table III.

Table IV shows the comparison between the optimal droop coefficients and corresponding real power dispatch of the 9 GFM IBRs in the two studied cases, compared to those with an initial set of droop coefficients. It can be seen that with relative similar total generation of 4.8 GW, the optimal solutions suggest to use different sets of droop coefficients for the IBRs to achieve the best congestion reduction in two congestion conditions in Cases 1 and 2. More importantly, the optimal droop is not necessary to be constantly proportional to the IBR ratings. For example, the droop coefficient of the GFM at Bus 32 is higher in Case 1 and smaller in Case 2 than that

TABLE III
TOTAL OVERLOADED POWER WITHOUT AND WITH OPTIMAL DROOP SETTINGS IN TWO CONGESTION CASES

Case	Without optimal droop	With optimal droop
Case 1	241 MW	62 MW (74% reduction)
Case 2	441 MW	207 MW (53% reduction)

TABLE IV
COMPARISON OF IBR GENERATION WITHOUT AND WITHOUT THE OPTIMAL DROOP SETTING

	Initial Case		Case 1 - Optimal		Case 2 - Optimal	
Bus	m^P	P^g	m^P	P^g	m^P	P^g
30	4.0%	375.3	1.0%	466.4	1.0%	480.0
31	3.0%	493.5	8,3%	446.9	10.0%	445.3
32	2.0%	621.8	10.0%	523.6	3.0%	575.1
33	2.0%	587.2	1.6%	589.8	10.0%	496.4
34	3.0%	436.9	10.0%	392.7	1.5%	488.9
35	3.0%	582.5	10.0%	523.6	4.3%	554.7
36	4.0%	476.9	10.0%	443.6	3.1%	486.7
37	3.0%	485.4	1.0%	583.0	1.0%	600.0
38	2.0%	768.6	1.0%	864.9	3.4%	701.5
	Total: 4,828MW		Total: 4,834MW		Total: 4,829MW	
	Frequency: 59.80Hz		Frequency: 59.84Hz		Frequency: 59.82Hz	

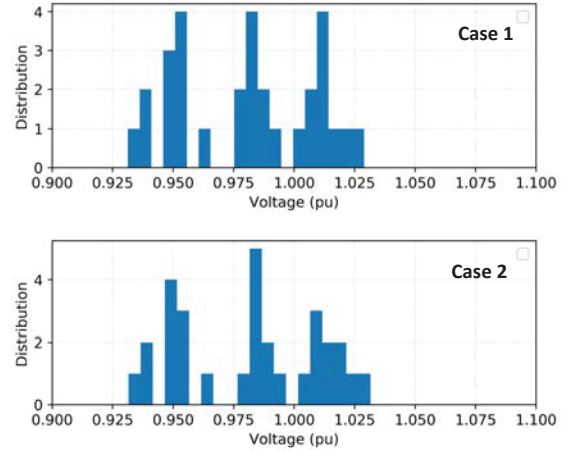


Fig. 5. Distribution of load voltages with optimal droop settings for GFM IBRs in Cases 1 and 2.

of the GFM at Bus 33. In addition, the optimal droop depends on the congestion condition in the system. For example, in Case 2 where Lines L5 and L8 are overloaded, the IBR at Bus 32 reduces the generation to 575.1 MW from 621.8 MW as in the initial case, in which overloading is not penalized.

The distributions of voltages at load buses in Cases 1 and 2 with the optimal droop settings are shown in Fig. 5. It is observed that these load voltages are within the required limits of 0.9 pu - 1.1 pu specified by the optimization input.

C. Verification of the Numerical Solution in EMT Simulation

Table V shows the comparison of bus voltages, frequency, line power, and GFM IBR generation from the numerical optimization solution and EMT simulation in PSCAD for Case 2. The two sets of results are nearly identical with marginal deviations. As an example, Fig. 6 shows the measured power generation from the 9 GFM IBRs in PSCAD. It can be seen that these simulation results are closely match the numerical values in Table IV.

TABLE V
COMPARISON OF NUMERICAL SOLUTION AND EMT SIMULATION RESULT

Metrics	Maximum deviation
Bus voltages	0.00037 pu
Steady-state frequency	0.001 Hz
GFM IBR power generation	0.6 MW or 0.001pu of M_{base}
Line power flow	0.4 MW

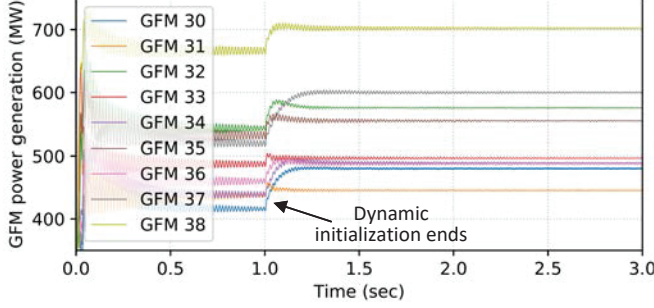


Fig. 6. Real power generation from 9 GFM IBRs in Case 2 from PSCAD.

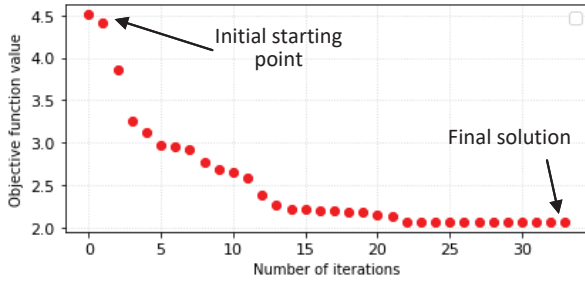


Fig. 7. Trajectory of the objective function in Case 2.

These similar results verifies the feasibility of the numerical solution and the accuracy of the proposed optimization model. More importantly, it confirms once again the efficacy of the optimal droop in reducing total overloaded line power while keeping other operational quantities such as voltage and frequency within required ranges.

D. Computational Performance

The trajectory of the objective function value in Case 2, which has more overloaded lines, is shown in Fig. 7. With the test IEEE 39-bus system, the solution converges after 33 iterations after 1.6 seconds. Therefore, the proposed formulation and solution method can be scaled for studying larger systems. On the other hand, the PSCAD simulation, which implements a switching GFM IBR model and uses a 5us time step, takes approximately 12 minutes to simulate 3 seconds, which is when the system can be considered of reaching steady state. Hence, the presented steady-state model also allows quick understanding about the steady-state status of an 100% GFM IBR-based power system without the need of running a lengthy dynamic simulation.

V. CONCLUSIONS AND FUTURE WORK

This paper proposes a steady-state optimization model for a power system with 100% droop-based GFM IBRs. The objective is to identify the optimal IBR droop coefficients to mitigate transmission congestion. The optimal solution shows that the GFM IBRs need to be set with different droop

coefficients under different congestion conditions instead of just being proportional to the IBR ratings as in conventional approaches. With the optimal settings, the total congestion in the test IEEE 39-bus system reduces 74% and 53% in two studied congestion conditions. Also, load voltage limits, frequency constraint, and IBR reserve requirements are met. The feasibility of the obtained numerical solution is verified by an EMT simulation with marginal deviations of all power flow quantities. Given these results and a significantly less computational time compared to the EMT simulation, future works will focus on large-scale systems with a mix of GFM and GFL IBRs and synchronous machines, considering uncertainties and variations of renewable generations and loads as well as solution solving time for real-time operation applications.

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